

Ragavendran S

Senior Full-Stack SDET | UI, API & Mobile Test Automation

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Summary

Full-stack SDET with 7+ years building scalable test frameworks and automation platforms across UI, API, and mobile ecosystems. Strong in Java, TypeScript, and Python with hands-on experience designing parallel test execution, retry mechanisms, and dynamic data injection. Led a Playwright migration reducing flakiness by 48% and CI time by 50%, built LLM-powered test generation tools cutting authoring effort by 60%, and drove automation coverage to 50–82% across 3 companies while mentoring 5+ engineers into independent contributors.

- Designed reusable framework modules in Java and TypeScript supporting parallel execution, retry logic, and synthetic test data injection, deployed across multiple product teams.
- Owned full-stack test coverage across UI (Playwright, Selenium), API (REST Assured, Karate), and mobile (Appium, Android/iOS), achieving 50–82% automation coverage across platforms and companies.
- Led QA strategy and framework rebuild across 3 companies, managing and upskilling teams of 5 engineers into independent contributors while coordinating automation direction with engineering leads.

Skills

Languages : Java • TypeScript • Python • JavaScript • SQL
UI Testing : Playwright • Selenium WebDriver • Cross-browser • Visual Regression • Accessibility Testing
API Testing : REST Assured • Karate • Postman • Contract Testing • Schema Validation
Mobile Testing : Appium (Android & iOS) • BrowserStack • Device Farm
Frameworks : Cucumber BDD • TestNG • Page Object Model • Allure Reports • JMeter
CI/CD & DevOps : Jenkins • GitHub Actions • GitLab CI • Azure DevOps • Docker • Maven • Git
AI & Tooling : FastAPI • GitHub Copilot • Windsurf • LLM Test Generation • Google Gemini • Vertex AI • BigQuery
Architecture : System Design • DSA • Design Patterns • Jira • Zephyr • Agile • Scrum

Experiences

iLink Digital, Chennai

Test Specialist

Jul 2025 – Present

- Reduced flaky tests by 48% and CI build time by 50% by migrating 200+ UI tests from Protractor to Playwright across 10 modules, validating critical screen flows, component interactions, and cross-browser rendering with visual regression checks.
- Expanded automation coverage to 82% (UI + API) by identifying and incorporating 20+ previously untested scenarios into regression suites, directly reducing defect volume reaching production validation.
- Cut manual verification by 40%/release by engineering Playwright POM-based validation flows with accessibility checks across login screens, multi-step onboarding forms, and core transactional pages.
- Configured parallel test execution with retry logic in TypeScript, reducing end-to-end test suite runtime by 35% and enabling faster feedback cycles on pull requests.
- Conducted performance profiling with JMeter for client reviews, identifying latency bottlenecks across critical API endpoints under concurrent load.

HCL Technologies Limited, Chennai

Technical Lead

Mar 2025 – Apr 2025

- Reduced manual test authoring by 60% by designing and deploying an LLM-powered test case generator (OpenAI API, FastAPI) that converts feature requirements into structured test scenarios, acceptance criteria, and automation skeletons.
- Designed QA automation playbooks and integration patterns for ML validation pipelines, producing a technical feasibility report with an adoption roadmap for reducing manual testing effort across future releases.

Comcast India Engineering Centre, Chennai

Software Development Engineer in Test – 3

Jun 2022 – Feb 2025

- Reduced incident triage time by 40% and partner coordination by 60% by building observability dashboards tracking pipeline failures, third-party vendor issues, and model regressions across a large-scale order fulfilment platform.
- Eliminated 90% of manual follow-ups for a 25-person cross-functional team by automating partner orchestration and validation workflows across ordering, tracking, and replacement journeys using Power Automate.
- Reduced compliance overhead by 30–50% by engineering JIRA JQL automation that standardized reporting workflows across multiple squads; earned Sky Stars Award from the platform's Assurance Lead.
- Validated data integrity across order fulfilment pipelines using SQL against production and staging databases, catching 15+ data transformation defects pre-release.
- Built synthetic test data pipelines using Python that generated validated datasets for integration and pipeline testing, removing backend dependency and unblocking release cycles for distributed teams.
- Optimized test execution pipelines by implementing intelligent test selection based on code change impact, reducing regression suite runtime by 30% while maintaining full coverage across critical paths.

- Designed and implemented a modular test framework in Java with parallel execution, configurable retry mechanisms, and dynamic test data injection, enabling independent execution across 3 Agile squads.
- Achieved 50% application coverage by refactoring legacy automation into a production-grade framework using Selenium WebDriver, REST Assured, Appium, TestNG and Cucumber with Page Object Model across 15+ service endpoints, integrated with Allure and Extent Reports in Jenkins.
- Implemented contract testing and schema validation across 15+ microservice endpoints to ensure backward compatibility and data consistency across distributed services.
- Scaled automation capability by upskilling 5 engineers from low familiarity to independent framework contributors, eliminating single points of expertise with test execution tracked in Zephyr.

Infosys Limited, Chennai

Test Engineer

Oct 2018 – Jun 2022

- Improved build feedback by 30% and reduced manual effort by 70% by owning Appium-based mobile test suites (Android + iOS) validated across device configurations on a large-scale transportation platform across multiple environments.
- Improved validation efficiency by 12% by implementing Karate API contract tests and log-based validation frameworks for backend data transformations, enabling faster defect resolution across distributed teams working across time zones.
- Investigated production issues by analyzing application logs, configuration files, and system behaviour, identifying root causes and providing development teams with reproducible findings and suggested fixes.
- Performed SQL-based backend validation verifying data consistency across ETL pipelines and service databases, flagging schema drift and transformation anomalies before release sign-off.

Senior Test Automation Executive

- Earned Rising Star Award by self-initiating the transition from manual to automated testing, independently learning and implementing UFT/QTP UI automation scripts on a safety-critical railway platform, directly reducing manual regression effort within the first year.
- Identified and reported 8% of total project defects with detailed JIRA documentation including reproduction steps, environment details, and suggested fixes, consistently accelerating development team resolution cycles.
- Created reusable test data templates and environment configuration scripts that standardized test setup across the team, reducing test preparation time by 25%.

Test Executive

- Executed manual functional testing and managed test suites using HP ALM across a safety-critical railway platform, documenting 18 verified defects with full traceability to requirements and test cases.
- Supported end-to-end STLC activities including test planning, test case design, execution, and closure, maintaining RTM coverage between business requirements and test execution results.
- Collaborated with cross-functional teams across requirements gathering, defect triage, and release sign-off cycles, ensuring comprehensive test coverage for safety-critical system updates.

Projects

AI-Based Test Case Generator

github.com/iamragavendrants/test-case-generator

Problem: Manual test authoring from feature requirements was slow and created traceability gaps.

Solution: Designed and deployed an LLM-powered system using OpenAI API and FastAPI that converts requirements into structured test scenarios.

Impact: Reduced manual test authoring effort by 60% and improved requirement-to-test traceability across the team.

Test Data Provider Tool

github.com/iamragavendrants/simple-test-data-generator

Problem: Teams depended on backend engineers for test data, creating delays and blocking release cycles.

Solution: Developed an internal platform generating validated synthetic datasets for integration, API, and pipeline testing.

Impact: Cut environment data wait time by 80% and eliminated backend dependency during release cycles.

Retention Rocket (Client-hosted Hackathon)

Problem: Retention decisions were being made on gut feel despite rich behavioral data being available.

Solution: Prototyped an AI-driven churn prediction platform using Python, Google Gemini, Vertex AI, and BigQuery on GCP.

Impact: Surfaced data-driven retention strategies and earned a Client Honorary Mention from the client panel.

Awards

2025 – **Client Honorary Mention:** Built Retention Rocket, an AI-driven churn prediction prototype, at a client-hosted hackathon.

2024 – **Sky Stars Award:** Awarded for JIRA JQL compliance automation that reduced team overhead by 30-50%.

2023 – **Champ Award:** Automated Excel-based metrics improving operational reporting efficiency across teams.

2022 – **First Time Right:** Delivered automation scripts for a new feature achieving UAT sign-off with near-zero defects.

2021 – **Rising Star:** Self-initiated automation on a safety-critical platform, exceeding expectations within the first year.

Certifications

- ISTQB® Certified Tester – Foundation Level
- Power BI – PwC Switzerland
- Data Science & Big Data Analytics – Dell EMC
- Innovation & Creativity – Harvard ManageMentor
- Python for Data Science – IBM

Education

Bachelor of Computer Application (BCA) Jul 2015 – Apr 2018
Kalasalingam Academy of Research and Education,
Krishnan Koil

- First Class with Distinction
- 2nd Place in Inter-college Product Innovation Competition